Higher National Diploma in Civil Engineering

First Year, Second Semester Examination – 2017

CE1205- Civil Engineering Construction I

Instructions for Candidates:	No. of questions	: 06
Answer any **Five (05)** questions.	No. of pages	: 02
All question carry equal marks	Time	: Three (03) hours

1.

a. A raft foundation depth of 2.5m to be excavated in ordinary soil, the site is surrounded by building structures in three sides. Identify the problems which may occur and suggest the method to carry out this work, including dealing with water, machinery and other safety issues. (6Marks)

b. Describe the following

i. Dead load (2Marks)

ii. Bearing capacity of soil. (2Marks)

iii. Plinth construction (2Marks)

iv. Hard core filling under the floor (2Marks)

c. Neat sketch diagram of isolated footing inclusive of all components (6Marks)

(Total 20 Marks)

2.

a. What is meant by 1:2:4 (20) concrete? (2Marks)

b. Briefly explain the purpose DPC for wall and DPM for ground floor. (4Marks)

c. Briefly explain the advantages and disadvantages of flat roof over pitch roof. (4Marks)

d. Briefly explain the procedure of rectifying a leakage in water supply in a house with necessary tools and materials (4Marks)

e. State any four types of paints used for different elements (4Marks)

f. State one advantage and one disadvantage of sea sand over river sand. (2Marks)

(Total 20 Marks)

3.

a. State the different between the solid slab and flat slab. (3Marks)

b. Define the "sewage" with example in a domestic building. (3Marks)

c. Briefly explain the following sewer systems and components
   i. Public sewer (2Marks)
   ii. Private sewer (2Marks)
   iii. Manhole (2Marks)
   iv. Septic tank (2Marks)
d. Discuss how an elevator differs from escalator. (3Marks)
e. Calculate the breaking capacity of Miniature Circuit Breaker (MCB) for an electric circuit which is designed to use an immersion heater 2500W and other appliances 1200W, use standard voltage in Sri Lanka (3Marks)

(Total 20 Marks)

4.
a. Write short notes on "Use of arch and construction process" (4Marks)
b. State requirements of external door. (3Marks)
c. State any four ironmongeries of doors and windows. (4Marks)
d. State any three types of windows in different method of operation (3Marks)
e. Briefly explain the advantage of system scaffolding. (3Marks)
f. Briefly explain the purpose of suspended scaffolding (3Marks)

(Total 20 Marks)

5.
a. Briefly explain the manufacture of cement (OPC). (6Marks)
b. Briefly explain the following elements in a building.
   i. Stiffener (3Marks)
   ii. Gable wall (3Marks)
   iii. Reveals (2Marks)
c. State any two appropriate equipment for the following tasks
   i. Excavation and earth moving (2Marks)
   ii. Transporting of materials (2Marks)
   iii. Compaction of soil (2Marks)

(Total 20 Marks)

6.
a. Briefly explain the importance of construction project management. (5Marks)
b. State any four documents maintained at a construction site (4Marks)
c. Briefly explain the quality control and quality assurance (6Marks)
d. What is a contract? (2Marks)
e. State types of contracts in a construction. (3Marks)

(Total 20 Marks)